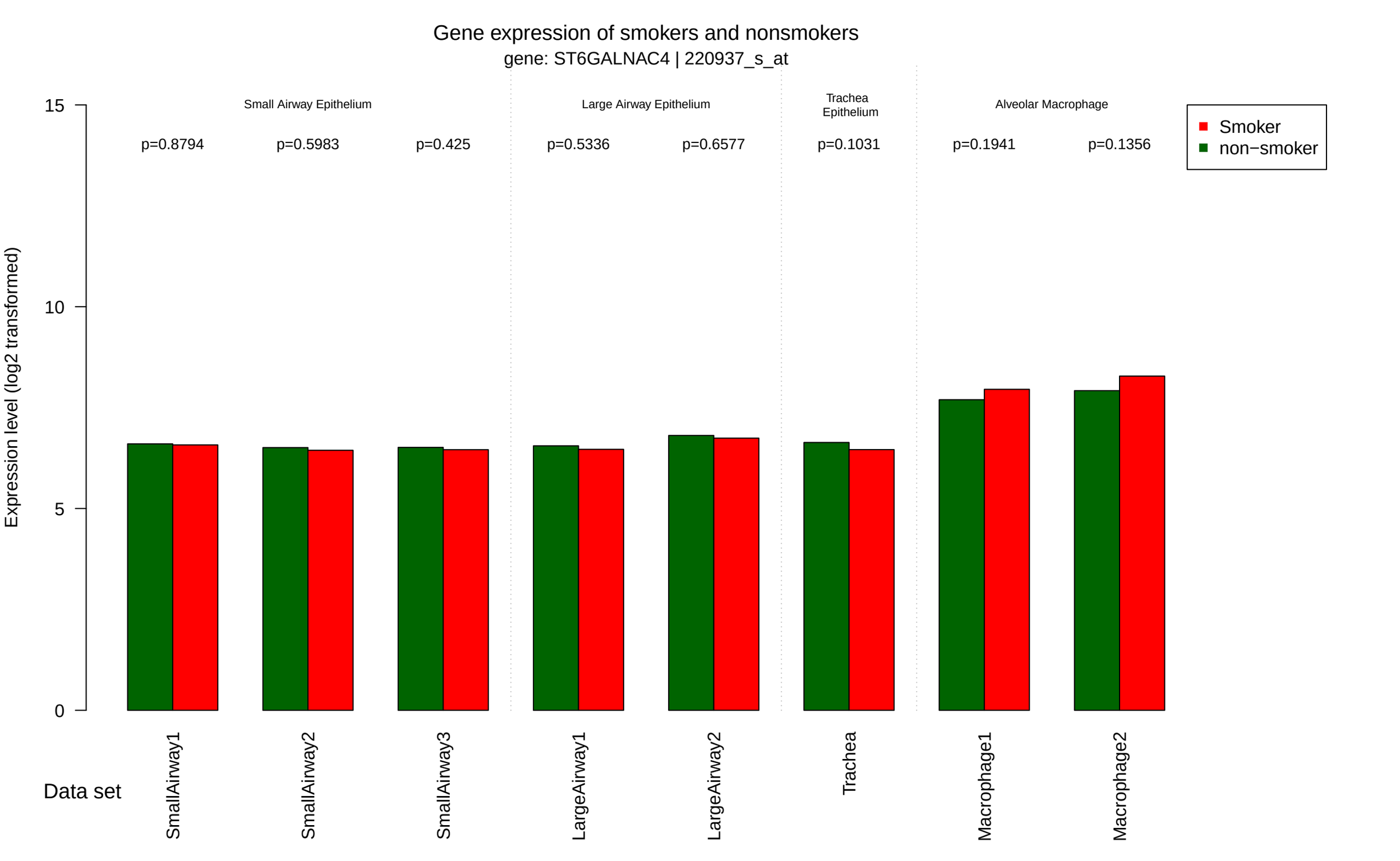
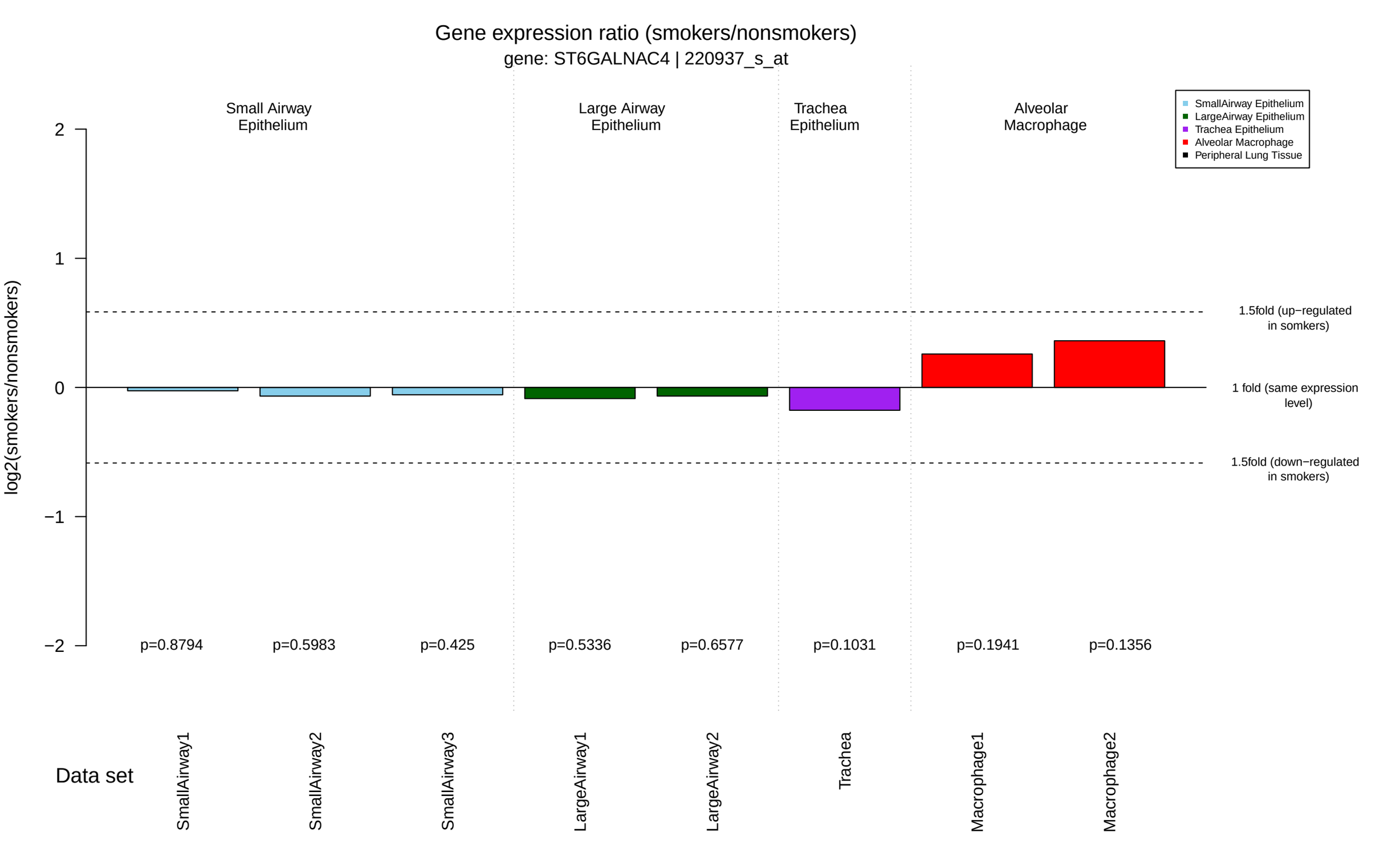
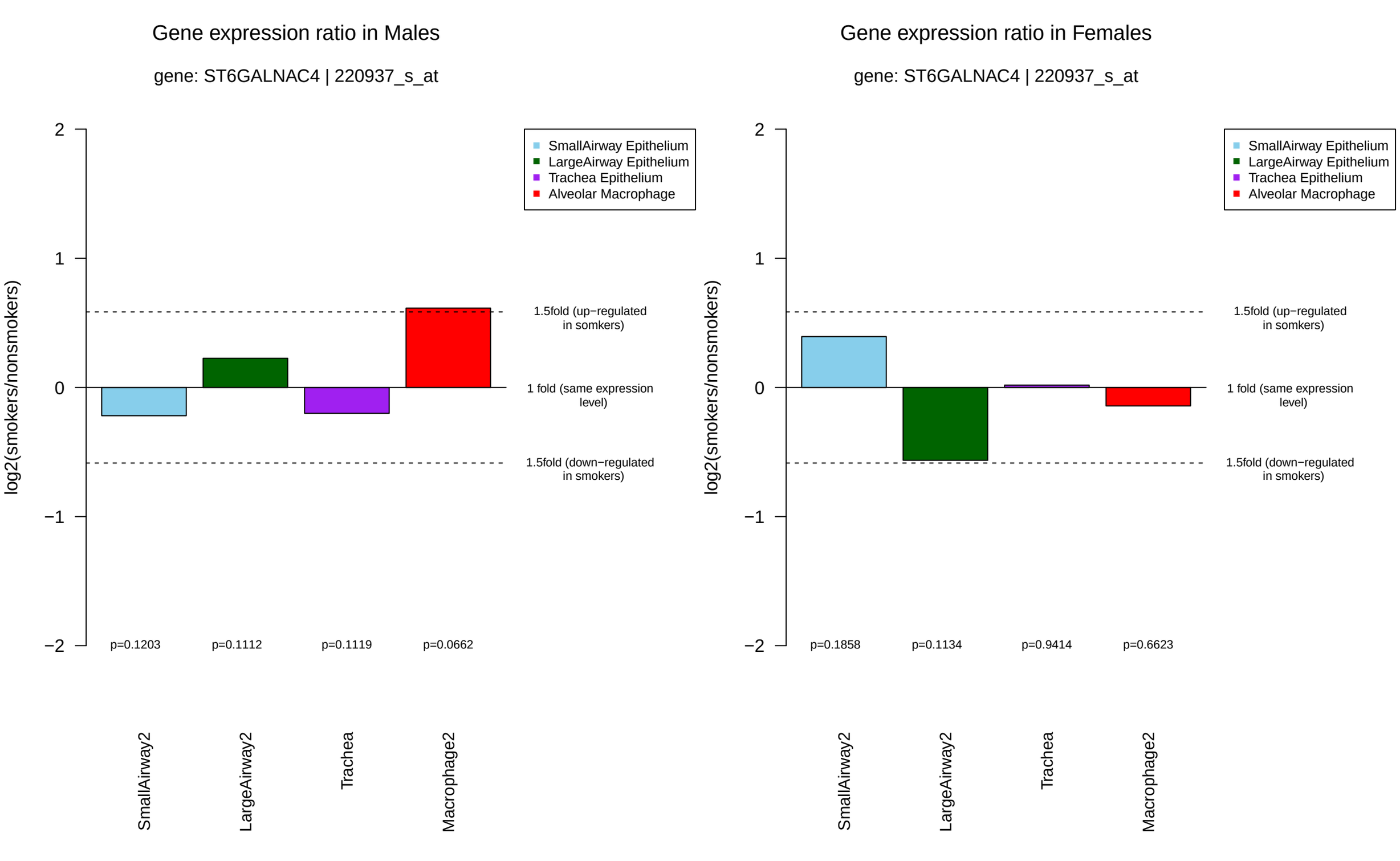
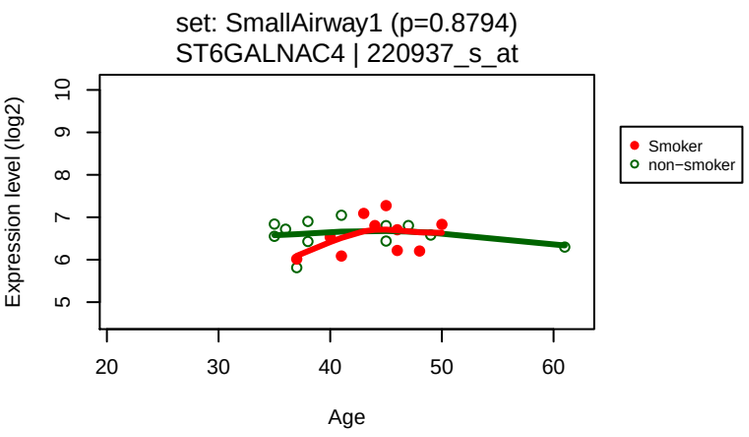
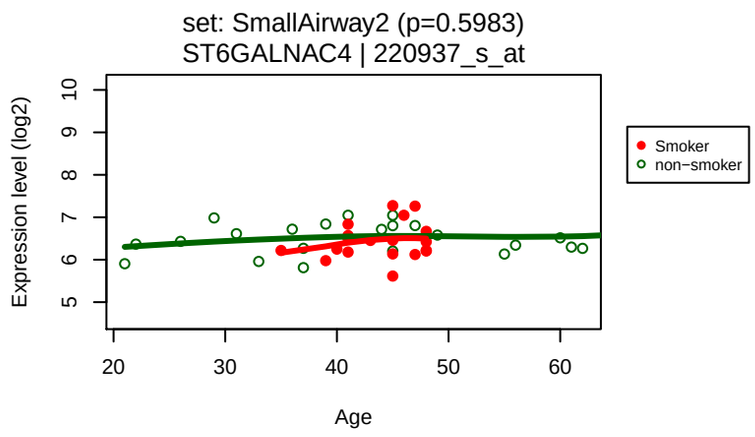
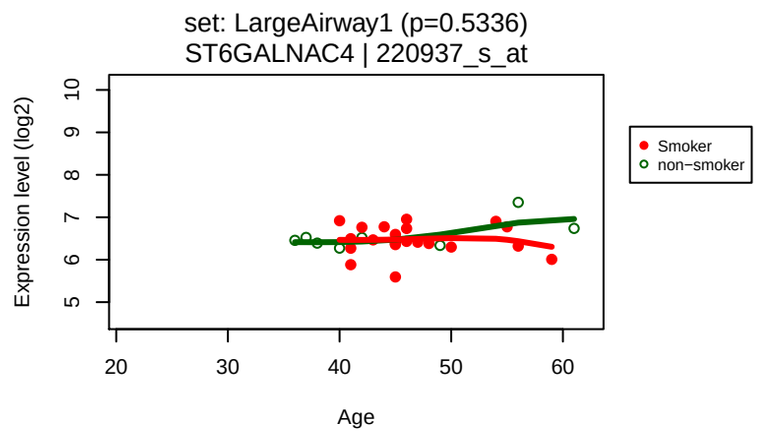

Gene expression of smokers and nonsmokers
gene: ST6GALNAC4 | 220937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8794
p=0.5983
p=0.425
p=0.5336
p=0.6577
p=0.1031
p=0.1941
p=0.1356
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

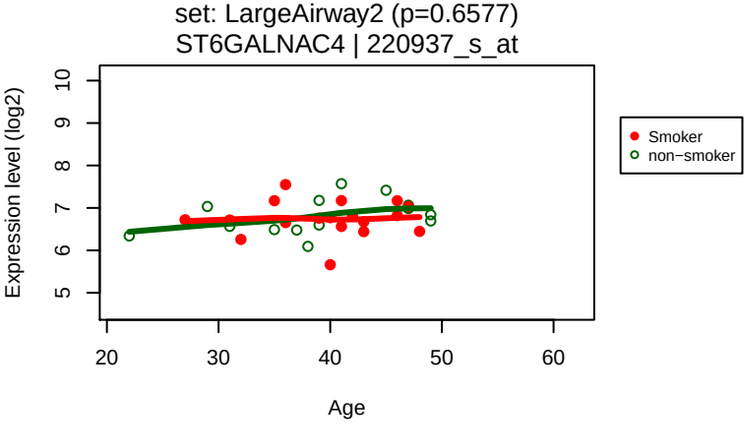

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC4 | 220937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8794
p=0.5983
p=0.425
p=0.5336
p=0.6577
p=0.1031
p=0.1941
p=0.1356
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

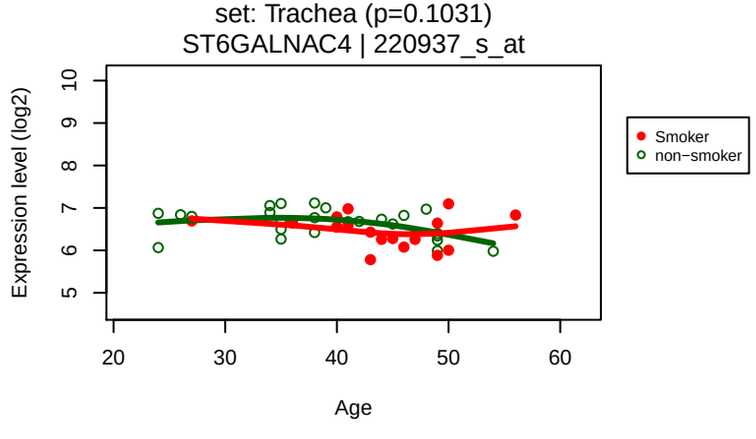

Gene expression ratio in Males
gene: ST6GALNAC4 | 220937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1203
p=0.1112
p=0.1119
p=0.0662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC4 | 220937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1858
p=0.1134
p=0.9414
p=0.6623
SmallAirway2
LargeAirway2
Trachea
Macrophage2

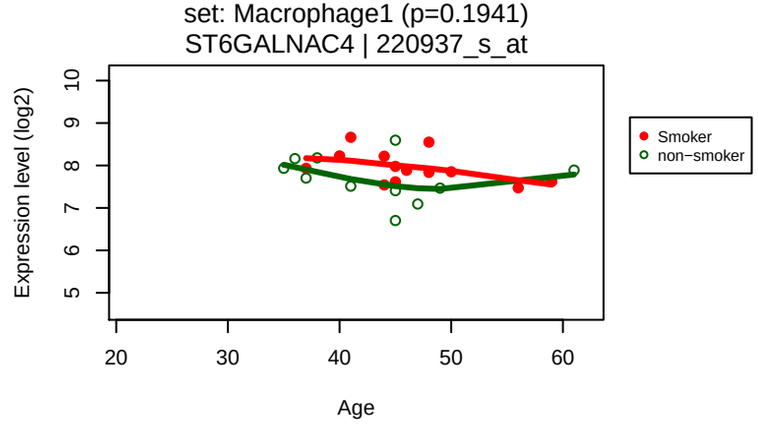

set: SmallAirway1 (p=0.8794)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker

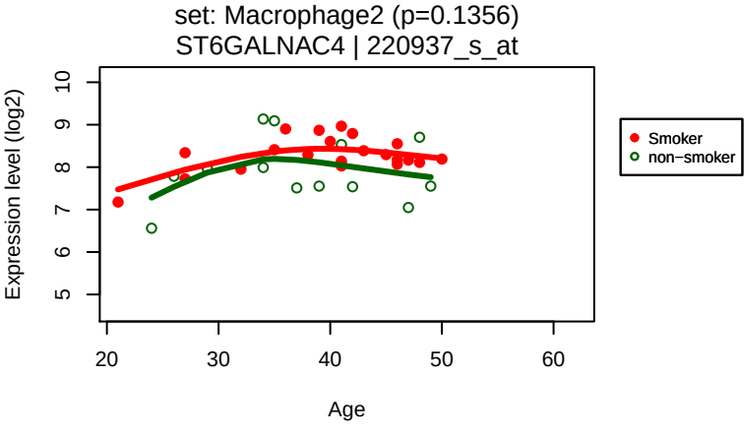

set: SmallAirway2 (p=0.5983)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker

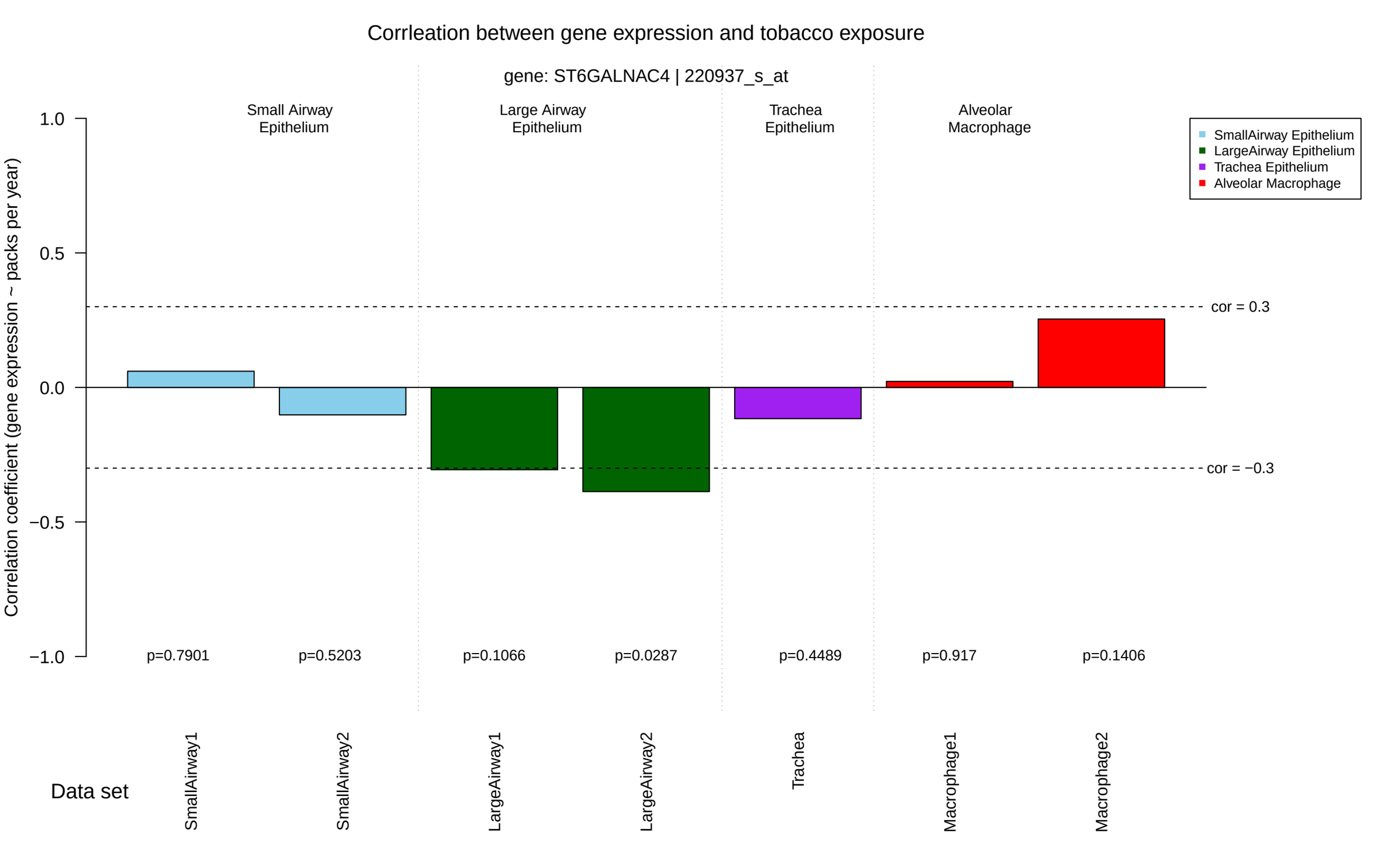

set: LargeAirway1 (p=0.5336)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6577)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1031)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1941)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1356)
ST6GALNAC4 | 220937_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC4 | 220937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7901
p=0.5203
p=0.1066
p=0.0287
p=0.4489
p=0.917
p=0.1406
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2